

Public Economics (ECON 131)

Section #11: Moral Hazard & Social Security

GSI: Steve Ramos

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1 Moral Hazard

1.1 Practice Problem

A firm has to decide an insurance plan to provide for a single worker. The worker has probability q equal to 0.2 of becoming unemployed, and $1 - q$ equal to 0.8 of staying employed. The worker can choose his unemployment duration, D , to be maximum 1 year, i.e. $D \in [0, 1]$.

If the worker stays employed he gets an income of $w = 100$ minus the insurance premium p . If he loses his job he gets the insurance benefit bD for the part of the year that he is unemployed, and $w(1 - D) = 100(1 - D)$ for the part of the year that he is employed. In addition, he gets disutility from searching for a job. Since searching less for a job implies longer unemployment durations, this is the same as saying that he gets positive utility from a longer unemployment duration, D . With an instantaneous utility function $u(c) = \ln(c)$ this means that expected utility is

$$EU = 0.8\ln(100 - p) + 0.2(\ln(bD + 100(1 - D) - p) + D)$$

where $\ln(100 - p)$ is his utility under employment and $\ln(bD + 100(1 - D) - p) + D$ is his utility under unemployment.

- (a) What is the benefit level as a function of the premium for an actuarially fair insurance program?

Solution: Under an actuarially fair program,

$$\text{Expected profits} = p - 0.2bD = 0$$

$$\text{so } p = 0.2bD \text{ and } b = \frac{p}{0.2D}$$

- (b) The insurance company expects that the worker will be unemployed for half a year on average, i.e. $D = 0.5$. Which premium and benefit level ensures full insurance under the actuarially fair insurance program in the previous question? Call this Policy #1.

Solution: For full insurance, consumption in both states are the same. In other words, consumer smooth their consumption so that their consumption when employed is equal to their consumption when unemployed. Therefore:

$$100 - p = bD + 100(1 - D) - p$$

$$100 = bD + 100 - 100D$$

$$100D = bD$$

$$b = 100$$

so the corresponding premium is $p = 0.2(100)(0.5) = 10$.

- (c) It turns out that the insurance company's guess about the unemployment duration length might not have been correct. Write up an expression for the worker's utility under Policy #1. Which D will the worker actually choose? What is his utility under Policy #1?

Solution:

$$EU = 0.8\ln(100 - p) + 0.2(\ln(bD + 100(1 - D) - p) + D)$$

$$EU = 0.8\ln(100 - 10) + 0.2\ln(100D + 100 - 100D - 10) + 0.2D$$

$$EU = 0.8\ln(90) + 0.2\ln(90) + 0.2D$$

$$EU = \ln(90) + 0.2D$$

Since utility is increasing in D , they will set D at its maximum, i.e. $D = 1$. Hence, his expected utility will be $\ln(90) + 0.2 = 4.6998$

However, if he remained at $D = 0.5$ and did not change his duration in response to being insured, his expected utility would be $\ln(90) + 0.2(0.5) = 4.5998$

- (d) The firm learns that it was losing money under Policy #1. Hence, it proposes a new actuarially fair insurance plan with full insurance under the actual D . Call this Policy #2. What is the new premium p and benefit b ? What is the worker's utility under this policy? Compare it to his utility under Policy #1. Would he choose to change his unemployment duration rather than keep it at $D = 0.5$?

Solution:

Under full insurance,

$$\begin{aligned}100 - p &= b(1) + 100(1 - 1) - p \\100 &= b\end{aligned}$$

so, $p = 0.2(100)(1) = 20$. His new expected utility will be:

$$EU = 0.8\ln(100 - 20) + 0.2\ln(100D + 100 - 100D - 20) + 0.2D$$

$$EU = 0.8\ln(100 - 20) + 0.2\ln(100(1) + 100 - 100(1) - 20) + 0.2(1)$$

$$EU = 0.8\ln(80) + 0.2\ln(80) + 0.2$$

$$EU = \ln(80) + 0.2$$

$$EU = 4.58$$

Therefore, he would rather have kept a duration of $D = 0.5$ and remained under Policy #1, getting an expected utility of 4.5998.

- (e) Now, suppose the firm implements a new plan (Policy #3) under which full insurance is not provided. In particular the firm provides a benefit of $b = 20$. Under this lower benefit the firm expects a low unemployment duration, namely $D = 0.25$. Under an actuarially fair program this means that $p = 1$ (check this!).

- (i) Which unemployment duration will the worker actually choose?

Solution: The worker's utility is now

$$EU = 0.8\ln(100 - 1) + 0.2\ln(20D + 100(1 - D) - 1) + 0.2D$$

$$EU = 0.8\ln(99) + 0.2\ln(99 - 80D) + 0.2D$$

Maximize with respect to D , such that

$$\frac{dEU}{dD} = \frac{0.2}{99 - 80D}(-80) + 0.2 = 0$$

$$0.2 = \frac{0.2}{99 - 80D}(80)$$

$$99 - 80D = 80$$

$$19 = 80D$$

$$D = \frac{19}{80} = 0.2375$$

- (ii) Will the firm lose money under this D ?

Solution: Because $\frac{19}{80} < 0.25$ the firm is actually making a profit.

- (iii) Compute the worker's utility. Which of the three policies does he prefer?

Solution: Under Policy #3, the worker's utility is:

$$EU = 0.8\ln(100 - 1) + 0.2\ln(20D + 100 - 100D - 1) + 0.2D$$

$$EU = 0.8\ln(99) + 0.2\ln(99 - 80D) + 0.2D$$

$$EU = 0.8\ln(99) + 0.2\ln(99 - 80(\frac{19}{80})) + 0.2(\frac{19}{80})$$

$$EU = 0.8\ln(99) + 0.2\ln(99 - 19) + 0.2(\frac{19}{80})$$

$$EU = 0.8\ln(99) + 0.2\ln(80) + 0.2(\frac{19}{80})$$

$$EU = 4.6$$

He will choose Policy #3!

- (iv) Why does your answer differ from the result in class that full insurance is optimal under no behavioral responses?

Solution: The reason why he won't choose full insurance is because under full insurance he can't commit to keeping a short unemployment duration. This implies that the insurance company will need to provide an expensive insurance plan in order to break even. However, under less-than-full insurance he will be able to commit to a low unemployment duration, so the insurance plan will be cheap, and he will get a higher utility.

2 Social Security

2.1 Definition and Motivation

- In the standard **life-cycle model**, rational individuals save while they work and consume savings when they retire
 - Uncertainty in returns to saving, life span, future ability to work, future health
 - Elderly would like *annuities*: guaranteed stream of income until end-of-life
- Rationales for govt intervention: individual failures, adverse selection in annuities market
 - For 2/3 of retirees, SS is > 50% of retirement *income* ($\approx 30\%$: >90% of income)
 - Retirees also have *wealth* (home ownership, employment pensions, extra savings), but often insufficient
- Government intervention to provide retirement programs has been common since the first part of the 20th century
 - **Funded Programs**: Workers contributions are invested in financial assets and will pay for benefits when they retire.
 - **Unfunded Program**: Benefits of current retirees are paid out of contributions from current workers.
- US Social Security's OASI is a mandated, unfunded annuity run by the federal government

2.2 Details

- Eligibility:
 - Funded by payroll taxes on workers (12.4% on first \$137,700 of earnings)
 - Need 10 years of contributions
- Payouts/benefits:
 - Nonlinear function of average earnings over highest 35 years (\$1400/month average)
 - Spouses can receive 50% of higher-earner's benefit ("spousal benefit")
 - Individuals can "claim" anytime after age 62, but benefits increase with your age at claiming (actuarially fair on average - severe distributional issues)
 - Wording is important/intentional: "reduced" benefits at 62-64 (**Early** Eligibility Age), "full" benefits at 65 (**Normal/Full** Benefits Age), mandated payments at 70
 - Beneficiaries are expected to be mostly-retired: increases in labor income reduce current benefits (beyond a small disregard; about 50% reduction on average)
 - **Social Security Wealth**: Expected PDV of Benefits - Expected PDV of payroll tax payments

2.3 Redistribution Practice Problem

What are the redistributive effects of social security?

Solution:

- Across generations: redistribution from young to old
- Within generations: redistribution from high- to low- income, but redistributive effects overstated if you ignore the fact that high-income people have higher expected life spans

2.4 Math Practice Problem

Consider the following model of social security. N people are born each period. Each person lives for two periods. In the first period of life, a person is young and in the second old. Thus, in any period after the initial period, half the population is young and half is old. Young people earn 2 chocolate bars while old people earn nothing. Assume that the chocolate melts, so there is no way for people to save privately.

Give a brief rationale for government provision of social security in this model.

Solution:

- Individuals cannot save privately since the chocolate melts so the government must provide social security so that they can eat when they are old.

Now suppose that the government of Candyland implements a social security system in the following manner. The government taxes each young person 1 chocolate bar and redistributes it to an old person in the same period. The program starts between periods 0 and 1 and ends between periods 2 and 3, as shown below. Let c_A^g denote the chocolate consumption of an agent from generation g (generation refers to the period the agent is born) at age A (age is either young or old).

(a) Under Candyland's social security program, fill in the blanks in the following chart:

Period 0	Period 1	Period 2	Period 3
Start SS		End SS	
$c_{young}^0 = 2$	$c_{old}^0 = 1$		
	$c_{young}^1 = 1$	$c_{old}^1 = 1$	
		$c_{young}^2 = 1$	$c_{old}^2 = 0$

Which generation gains the most in terms of consumption? The least?

Solution: Generation 0 benefits most, being able to enjoy the benefits of the social security system without having to incur the costs of paying for a previous generation's consumption. On the other hand, generation 2 loses the most, since they pay the tax but never enjoy the benefits of the system.

- (b) What assumption does this model make about the effect of social security provision on retirement behavior? Discuss how empirical evidence on retirement decisions and social security relates to this assumption.

Solution: The model assumes that the fraction of people who are "old" is invariant to the social security system. In other words, it assumes that retirement behavior is not affected by the provision of SS. In contrast, Gruber and Wise (2002) present evidence on retirement behavior around the world, and find that workers's retirement decisions are heavily influenced by whether or not additional work will increase their social security benefits.

From lecture: Gruber and Wise (1999) calculated the implicit tax (on work) from Social Security for a series of countries.

- Across countries, there is a great deal of variation in the implicit tax rate.
- Implicit tax close to zero for 62-year-olds in the United States.
- 91% in the Netherlands.
- And countries with higher taxes have less elderly labor force participation.

- (c) The country of Twixland is also considering implementing social security. However, the ingenious residents of Twixland have figured out a way to freeze and save chocolate for retirement. All of them have utility over consumption when young (c_A) and old (c_B) given by $\log(c_A) + \log(c_B)$. The demographic and other aspects of the economy are as in Candyland.

Replicate the chart in (b) for Twixland, assuming that the start and end of SS are completely unanticipated by its residents.

Solution: Since the utility is concave, individuals will want to smooth consumption perfectly across generations consume 1 chocolate bar when they are young and 1 when they are old. In Twixland, this is feasible in the absence of SS because people can save.

Twixlanders react to the new SS program by changing their savings decisions accordingly if they anticipate the receipt of SS when they are old. Generation 0 individuals do not anticipate getting SS when old, so they continue to save 1 chocolate bar in period 0, but gain an extra chocolate bar in period 1 because of SS. Generation 1 anticipates SS, so its consumption is unchanged (100% crowdout of government intervention). Generation 2 anticipates getting SS and therefore does not save for itself, but then gets nothing in period 3.

Period 0	Period 1	Period 2	Period 3
	Start SS		End SS
$c_{young}^0 = 1$	$c_{old}^0 = 2$		
	$c_{young}^1 = 1$	$c_{old}^1 = 1$	
		$c_{young}^2 = 1$	$c_{old}^2 = 0$

Again, we see that generation 0 benefits the most and generation 2 loses. The only difference relative to CandyLand is that generation 0 gains more consumption when old rather than young.